

```

import 'package:flutter/material.dart';
import 'dart:math';

void main() {
  runApp(const MortgageApp());
}

class MortgageApp extends StatelessWidget {
  const MortgageApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Mortgage Calculator',
      home: const MortgageHomePage(),
    );
  }
}

// Model class
class Mortgage {
  double amount = 100000;
  int years = 30;
  double rate = 0.035; // 3.5%

  double monthlyPayment() {
    int numberOfPayments = years * 12;

    if (numberOfPayments == 0) {
      return 0;
    }

    if (rate == 0) {
      return amount / numberOfPayments;
    }

    double monthlyRate = rate / 12;

    return amount * monthlyRate / (1 - (1 / pow(1 + monthlyRate, numberOfPayments)));
  }

  double totalPayment() {
    return monthlyPayment() * years * 12;
  }
}

class MortgageHomePage extends StatefulWidget {
  const MortgageHomePage({super.key});

  @override
  State<MortgageHomePage> createState() => _MortgageHomePageState();
}

class _MortgageHomePageState extends State<MortgageHomePage> {
  final Mortgage mortgage = Mortgage();

  bool termsAccepted = false;

  String money(double value) {
    return '\${value.toStringAsFixed(2)}';
  }
}

```

```

void openModifyScreen() async {
  await Navigator.push(
    context,
    MaterialPageRoute(
      builder: (context) {
        return ModifyScreen(mortgage: mortgage);
      },
    ),
  );

  setState(() {});
}

void showTermsDialog(bool? value) async {
  if (value != true) {
    setState(() {
      termsAccepted = false;
    });
    return;
  }

  bool? accepted = await showDialog<bool>(
    context: context,
    builder: (context) {
      return AlertDialog(
        title: const Text('Terms and Conditions'),
        content: const Text(
          'Do you agree to the terms and conditions?',
        ),
        actions: [
          TextButton(
            onPressed: () {
              Navigator.pop(context, false);
            },
            child: const Text('Cancel'),
          ),
          ElevatedButton(
            onPressed: () {
              Navigator.pop(context, true);
            },
            child: const Text('Agree'),
          ),
        ],
      );
    },
  );

  setState(() {
    termsAccepted = accepted == true;
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Mortgage Calculator'),
      backgroundColor: Colors.green[100],
    ),
    body: Padding(
      padding: const EdgeInsets.all(20),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,

```

```

children: [
  const Text(
    'Mortgage Information',
    style: TextStyle(
      fontSize: 24,
      fontWeight: FontWeight.bold,
    ),
  ),

  const SizedBox(height: 25),

  Text(
    'Amount: ${money(mortgage.amount)}',
    style: const TextStyle(fontSize: 18),
  ),

  const SizedBox(height: 15),

  Text(
    'Years: ${mortgage.years}',
    style: const TextStyle(fontSize: 18),
  ),

  const SizedBox(height: 15),

  Text(
    'Interest Rate: ${(mortgage.rate * 100).toStringAsFixed(2)}%',
    style: const TextStyle(fontSize: 18),
  ),

  const SizedBox(height: 15),

  Text(
    'Monthly Payment: ${money(mortgage.monthlyPayment())}',
    style: const TextStyle(fontSize: 18),
  ),

  const SizedBox(height: 15),

  Text(
    'Total Payment: ${money(mortgage.totalPayment())}',
    style: const TextStyle(fontSize: 18),
  ),

  const SizedBox(height: 10),

  CheckboxListTile(
    contentPadding: EdgeInsets.zero,
    title: const Text('Terms and Conditions'),
    value: termsAccepted,
    onChanged: showTermsDialog,
  ),

  const Spacer(),

  SizedBox(
    width: double.infinity,
    child: ElevatedButton(
      onPressed: openModifyScreen,
      child: const Text('Modify'),
    ),
  ),
],

```

```

    ),
  ),
);
}
}

class ModifyScreen extends StatefulWidget {
  final Mortgage mortgage;

  const ModifyScreen({
    super.key,
    required this.mortgage,
  });

  @override
  State<ModifyScreen> createState() => _ModifyScreenState();
}

class _ModifyScreenState extends State<ModifyScreen> {
  late TextEditingController amountController;
  late TextEditingController yearsController;

  late double selectedRate;

  @override
  void initState() {
    super.initState();

    amountController = TextEditingController(
      text: widget.mortgage.amount.toString(),
    );

    yearsController = TextEditingController(
      text: widget.mortgage.years.toString(),
    );

    selectedRate = widget.mortgage.rate;
  }

  @override
  void dispose() {
    amountController.dispose();
    yearsController.dispose();
    super.dispose();
  }

  void saveChanges() {
    double? amount = double.tryParse(amountController.text);
    int? years = int.tryParse(yearsController.text);

    if (amount == null || years == null || amount < 0 || years < 0) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text('Please enter valid values.'),
        ),
      );
      return;
    }

    widget.mortgage.amount = amount;
    widget.mortgage.years = years;
    widget.mortgage.rate = selectedRate;
  }
}

```

```

Navigator.pop(context);
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Modify Mortgage'),
    ),
    body: Padding(
      padding: const EdgeInsets.all(20),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          TextField(
            controller: amountController,
            keyboardType: TextInputType.number,
            decoration: const InputDecoration(
              labelText: 'Amount',
              border: OutlineInputBorder(),
            ),
          ),

          const SizedBox(height: 15),

          TextField(
            controller: yearsController,
            keyboardType: TextInputType.number,
            decoration: const InputDecoration(
              labelText: 'Number of Years',
              border: OutlineInputBorder(),
            ),
          ),

          const SizedBox(height: 20),

          const Text(
            'Select Interest Rate',
            style: TextStyle(
              fontSize: 18,
              fontWeight: FontWeight.bold,
            ),
          ),

          const SizedBox(height: 10),

          Expanded(
            child: ListView.builder(
              itemCount: 53,
              itemBuilder: (context, index) {
                double ratePercent = 2 + (index * 0.25);
                double rateDecimal = ratePercent / 100;

                return RadioListTile<double>(
                  title: Text(
                    '${ratePercent.toStringAsFixed(2)}%',
                  ),
                  value: rateDecimal,
                  groupValue: selectedRate,
                  onChanged: (value) {
                    setState(() {
                      selectedRate = value!;
                    });
                  },
                );
              },
            ),
          ),
        ],
      ),
    ),
  );
}

```

```
    },
  );
},
),
),

  SizedBox(
    width: double.infinity,
    child: ElevatedButton(
      onPressed: saveChanges,
      child: const Text('Done'),
    ),
  ),
),
),
),
),
);
}
```